

GI73HW01.002 ECN

PGDBC0

2022/01/19

ECN Notice



- Purpose:
 - Because the MegaChips Corporation (MCC) TCON business group will be closed, AUO Display Plus (ADP) will replace MCC TCON IC I2303 K02 with Raydium TCON IC I2303 Z09 for this model.
 - And LED driver IC RT8488GQW is going to EOL too, ADP will replace LED driver IC RT8488GQW with NT50508B for this model.
- Changes:
 - TCON IC will be changed from MCC I2303 K02 to Raydium I2303 Z09.
 - LED driver IC will be changed from RT8488GQW to NT50508B.
- Change Items:
 - PCBA TCON IC, LED driver IC
- Schedule

Sample delivery	AUO RA test completed	Target EC date	Phase-in
Jan 2022	Dec 2021 TCON IC and LED driver RA PASS	Mar 2022	Apr 2022

Change point



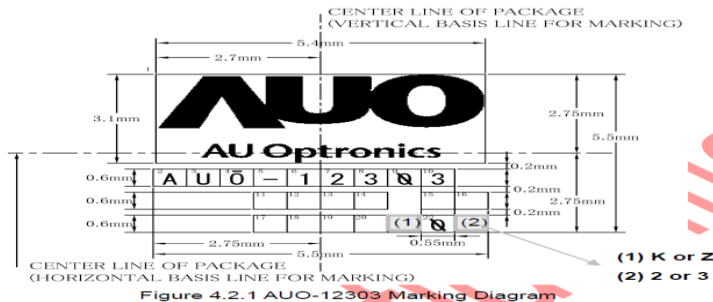
Original

AUO I2303 K02

New

AUO I2303 Z09

(Second line last number is 9)



AUO Display+

Change point

Original
RT8488GQW

New
NT 50508B



Evaluation result



Summary items and result

1. Optical measurement result: Pass
2. ESD/ EMI Result: Pass
3. RA Test: Pass

Evaluation result

Conclusion

For G173HW01.002 new Tcon IC verification, there is no changes in electrical characteristics, outer shape, and other contents described in the product spec.

However LED driver IC EC will change dimming duty cycle from 5% (200~20KHz) to 5% in low frequency range 200~5KHz(I) and 15% in high frequency range 5K~20KHz respectively.

F_{PWM}	Dimming Frequency	200	-	20K	[Hz]
	Swing Voltage	3	3.3	5	[Volt]
	Dimming duty cycle	5	-	100	%



F_{PWM}	Dimming Frequency	200	-	20k	[Hz]
	Swing Voltage	3	3.3	5	V
	Dimming Duty 200~5K	5	-	100	%
	Dimming Duty 5K~20K	15	-	100	%

Evaluation result



Optical Result: Pass

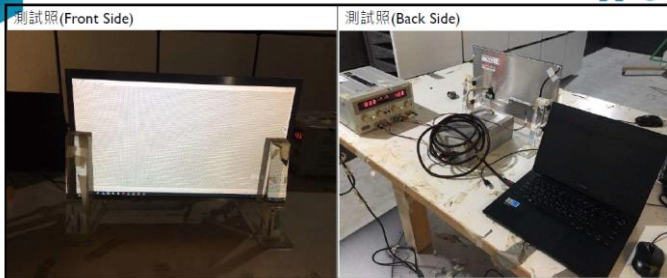
Originator	chongfeng				Model/PCS		G173HW01 V0/5PCS			Date	2021/8/29		
Test Condition	Lum.,CR,Chromaticity,Uniformity,Gary Scale,V-T,View Angle,RT,CrossTalk,Flicker												
Test System	Detector		Inverter			DC Voltage					Warm up time		
FPM-520-2	SR-UL1R		NA			12V					30 min		
【Test Reason】													
G173HW01.0 T-CON验证													
【Judgement】													
PASS													
Test Items	unit	PSS			23.6℃	23.6℃	23.6℃	23.6℃	23.6℃	-4σ	mean	+4σ	
					2021/8/24 22:05	2021/8/24 22:49	2021/8/24 23:20	2021/8/24 23:50	2021/8/24 0:34				
		2153DD6WLZZS010	205AMF9NCZZS010	2153807HJZZS010	205AMG2SFZZS010	2153805ABZZS010							
		min	typ	max									
Center Brightness	cd/m ²	320				420.3	364.3	435.8	419.1	438.1	295.849	415.5	535.191
Darkness (L0)						0.59	0.53	0.54	0.55	0.59	0.442	0.56	0.676
Contrast Ratio(CR)	---	450	600			713.9	689.3	810.3	763.4	740.3	557.085	743.5	929.822
Brightness(5P Ave.)	cd/m ²					405.3	356.6	424.5	411.7	425.2	291.869	404.7	517.451
Chromaticity													
	Wx	---	0.254	0.304	0.354	0.307	0.297	0.300	0.300	0.304	0.286	0.302	0.317
	Wy	---	0.263	0.313	0.363	0.319	0.296	0.307	0.304	0.309	0.273	0.307	0.341
	Rx	---	0.590	0.640	0.690	0.641	0.638	0.641	0.640	0.641	0.635	0.640	0.646
	Ry	---	0.296	0.346	0.396	0.343	0.341	0.343	0.342	0.342	0.339	0.342	0.345
	Gx	---	0.265	0.315	0.365	0.320	0.320	0.318	0.319	0.322	0.314	0.320	0.326
	Gy	---	0.571	0.621	0.671	0.620	0.622	0.619	0.619	0.624	0.611	0.621	0.630
	Bx	---	0.098	0.148	0.198	0.150	0.153	0.149	0.150	0.153	0.143	0.151	0.159
	By	---	0.004	0.054	0.104	0.054	0.046	0.052	0.050	0.048	0.037	0.050	0.062
	NTSC	---	---	---	---	72.3	72.7	72.7	72.3	73.2	71.232	72.655	74.078
	CCT	---	---	---	---	6932	8037	7605	7630	7253	5819.135	7491.607	9164.080

AUO Display+

Evaluation result

EMI Result: Pass

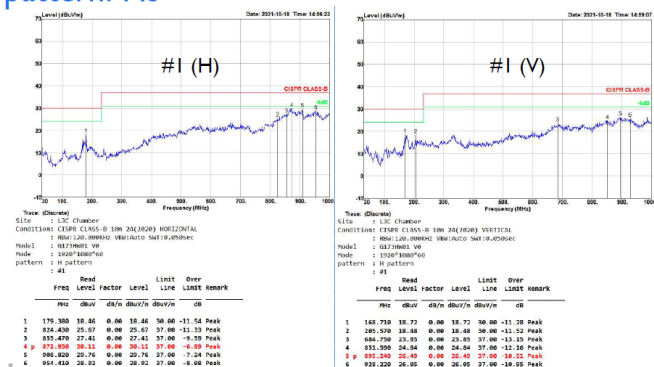
GI73HW01 V0 Test Condition



Test Equipment : R&S EMI Receiver : ESR07 (Cal. No. : F-03-01-0002 : Exp. : 2021/1/16)

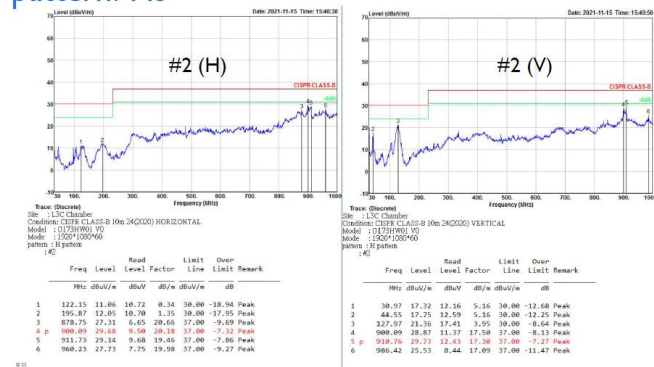
GI73HW01 V0 EMI test Report pattern: Hs

LIC chamber 1920*1080*60 patternGen ASIC: AUO-12303 Z09 XJC: NT65706 YJC: AU65207CH-D4EA



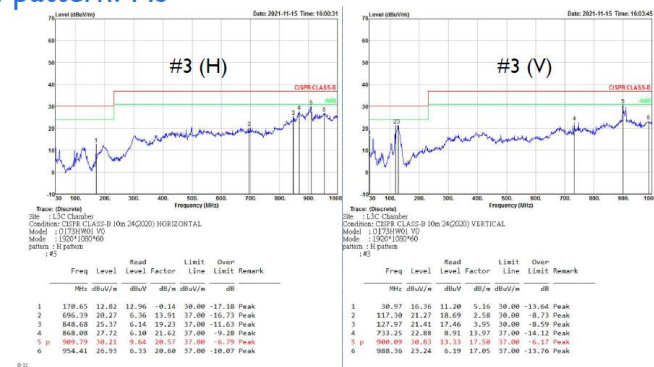
GI73HW01 V0 EMI test Report pattern: Hs

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GI73HW01 V0 EMI test Report pattern: Hs

LIC chamber 1920*1080*60 patternGen ASIC: AUO-12303 Z09 XJC: NT65706 YJC: AU65207CH-D4EA



Evaluation result



ESD Result: Pass

☐ Record ☒ ESD check ☐ Customer check

File Name	20210923-5D2190049-G173HW01 V0			Test Ambient: 26 °C / 56% RH											
Application Sec.	PRSGA4	User	朱嘉昇	Ext.	8535-3222	Sample Received	2021/9/27								
Model Name	G173HW01 V0	Quantity (pcs)	3	Panel S/N	G205AMG5BLZZ-ZS0102	(#1)									
					G205AMG5XCZZ-ZS0102	(#2)									
					G2153808UEZZ-ZS0102	(#3)									
Examiner	汪佳靜	Ext.	4202	Date Tested	2021/9/28	Date Completed	2021/9/28								
Test standard	☒ AUO Fixed Point Position Rules ☐ AUO IC Nearly Point Position Rules ☐ Customer Rules														
Stage	DVT		Resolution	Frame rate (Hz)	Color depth (bit)	Pixel clock (MHz)	Level B								
			1920 X 1080	60	8	75									
XPCB Part №		Layout №		PCB Vendor	健鼎										
YPCB Part №		Layout №		PCBA Vendor	台表										
C/B Part №	55.17G03.C04	Layout №	17G03-C04	ASIC	AUO-12303 Z09										
Inverter Part №	NA	Inverter Vendor	NA	SOURCE IC	46.S1026.198										
				GATE IC	46.G0900.021										
Purposes	DVT , G173HW01 V0														
Test Specification: Test Level Requirement: B					Results PASS										
Test Standard: AUO Fixed Point Position Rules (8 points, 25 times/point)															
Acceptance Definition:															
<table><tr><td>Class A</td><td>Normal operation. No degradation. No failures.</td></tr><tr><td>Class B</td><td>Some performance degradation allowed.</td></tr><tr><td>Class C</td><td>Temporary performance degradation. Recovery by operator (Reset) is acceptable. No Hardware failures.</td></tr><tr><td>Class D</td><td>Permanent damage.</td></tr></table>								Class A	Normal operation. No degradation. No failures.	Class B	Some performance degradation allowed.	Class C	Temporary performance degradation. Recovery by operator (Reset) is acceptable. No Hardware failures.	Class D	Permanent damage.
Class A	Normal operation. No degradation. No failures.														
Class B	Some performance degradation allowed.														
Class C	Temporary performance degradation. Recovery by operator (Reset) is acceptable. No Hardware failures.														
Class D	Permanent damage.														

Display+

Evaluation result

RA Result: Pass

Product No	G173H-W01 V0		Part No	97.17G03.002		Applicant/Dept		吴喜华/IRGMRB4		Judgement	
Test Purpose	G173HW01.0 T-CON验证					Application Date		2021/08/26		PASS	
RA No	R210827020		Product Stage		DVT		Input Date		2021/08/30		
							Issue Date		2021/09/01		
Test Item	Test Condition	Test Panel				Test Result					
Temperature Humidity Bias	50°C 80% 300h	No	Chip_Id	Shipping_No	T0★	T100★	T200★	T300★	Judgement		
		R2108270200001		G2153DD7DCZZS0102	OK	OK	OK				
		R2108270200002		G2153DD7DBZZS0102	OK	OK	OK				
		R2108270200003		G2153806MDZZS0102	OK	OK	OK				
Hot-start	70°C 5	No	Chip_Id	Shipping_No	T0★	T1+F5★			Judgement		
		R2108270200004		G2153DD8WBZZS0102	OK	OK			PASS		
		R2108270200005		G2153806WJZZS0102	OK	OK			PASS		
		R2108270200006		G2153DD7FMZZS0102	OK	OK			PASS		
High Temperature Operation	70°C 300h	No	Chip_Id	Shipping_No	T0★	T100★	T200★	T300★	Judgement		
		R2108270200004		G2153DD8WBZZS0102	OK	OK	OK	OK	PASS		
		R2108270200005		G2153806WJZZS0102	OK	OK	OK	OK	PASS		
		R2108270200006		G2153DD7FMZZS0102	OK	OK	OK	OK	PASS		
Cold-start	0°C 5	No	Chip_Id	Shipping_No	T0★	T1+F5★			Judgement		
		R2108270200007		G2153DD6WEZZS0102	OK	OK			PASS		
		R2108270200008		G2153DD6WGZZS0102	OK	OK			PASS		
		R2108270200009		G205AMG4EKZZS0102	OK	OK			PASS		
Low Temperature Operation	0°C 300h	No	Chip_Id	Shipping_No	T0★	T100★	T200★	T300★	Judgement		
		R2108270200007		G2153DD6WEZZS0102	OK	OK	OK	OK	PASS		
		R2108270200008		G2153DD6WGZZS0102	OK	OK	OK	OK	PASS		
		R2108270200010		G205AMG4EKZZZ@	OK	OK	OK	OK	PASS		
High Temperature Storage	70°C 300h	No	Chip_Id	Shipping_No	T0★	T100★	T200★	T300★	Judgement		
		R2108270200011		G2153DD8WCZZS0102	OK	OK	OK	OK	PASS		
		R2108270200012		G2153DD6WAZZS0102	OK	OK	OK	OK	PASS		
		R2108270200013		G2153DD7VLZZS0102	OK	OK	OK	OK	PASS		
Low Temperature Storage	-20°C 300h	No	Chip_Id	Shipping_No	T0★	T100★	T200★	T300★	Judgement		
		R2108270200014		G205AMG2CDZZS0102	OK	OK	OK	OK	PASS		
		R2108270200015		G2153806XJZZS0102	OK	OK	OK	OK	PASS		
		R2108270200016		G2153DD8WAZZS0102	OK	OK	OK	OK	PASS		
Thermal Shock Test	-20~60°C (-20°C (0.5h)~60°C(0.5h) 100cycles	No	Chip_Id	Shipping_No	C0★	C100★			Judgement		
		R2108270200017		G2153DD6WFZZS0102	OK	OK			PASS		
		R2108270200018		G2153DD7DAZZS0102	OK	OK			PASS		
		R2108270200019		G2153DD6WCZZS0102	OK	OK			PASS		

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